PulseCRM — AI-Native Customer Engagement Platform

Product Design Document

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1. Problem Statement

Modern CRM systems provide marketers with large amounts of customer data but very little guidance on what action should be taken. Marketers are expected to manually analyze customer behavior, identify potential customer segments, decide whom to target, create campaign content, launch campaigns, and finally analyze campaign performance.

This process is:

- Time consuming
- Data intensive
- Reactive instead of proactive
- Difficult for non-technical marketers

The objective of PulseCRM is to transform a traditional CRM into an **AI-first decision support system** where AI continuously analyzes customer behavior, recommends actionable customer segments, assists in campaign creation, and measures campaign effectiveness.

The marketer should spend time making business decisions—not manually searching through thousands of customers.

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2. Design Philosophy

Traditional CRM

Customer Data

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Manual Analysis

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Manual Segmentation

↓

Manual Campaign

↓

Analytics

PulseCRM

Customer Data

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AI Analysis

↓

Business Opportunities

↓

Marketer Decision

↓

Campaign Execution

↓

Performance Analytics

The system recommends.

The marketer decides.

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3. Product Goals

The application should enable marketers to:

- Import customer and order datasets
- Automatically analyze customer behavior

- Discover high-value customer opportunities
- Build campaigns with AI assistance
- Preview campaign communication
- Simulate campaign execution
- Measure communication performance
- Attribute revenue generated from campaigns

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4. Core Workflow

The application follows one continuous workflow.

Import Dataset

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Analyze Customers

↓

Generate Customer Segments

↓

Review Segment Intelligence

↓

Create Campaign

↓

Generate AI Message

↓

Preview Communication

↓

Launch Campaign

↓

Track Performance

↓

Campaign Analytics

Every page exists to support this workflow.

No page should work independently.

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5. Application Stages

Stage 1 — Data Import

Purpose

Import customer and order datasets.

Features

- Upload CSV
- Dataset validation
- Customer parsing
- Order parsing
- Duplicate handling
- Merge or Replace dataset

Output

Customer records stored in MongoDB.

Orders stored in MongoDB.

Automatically trigger ML analysis.

Stage 2 — Customer Intelligence

Purpose

Convert raw customer data into business insights.

Machine Learning Pipeline

Customer Orders

↓

RFM Feature Engineering

↓

Feature Scaling

↓

KMeans Clustering

↓

Business Personas

↓

Segment Analytics

Output

Every customer receives a business persona.

Example

- Ghost Diners
- Premium Diners
- Weekend Warriors
- Night Snackers
- Morning Regulars

These personas drive every future recommendation.

Stage 3 — Dashboard Intelligence

Purpose

Provide actionable business opportunities.

The dashboard does not display raw customer information.

Instead it answers:

"What should I do today?"

Dashboard Components

- Dataset status
- AI Opportunities
- Business KPIs
- Running campaigns
- Recent campaign performance

Dashboard adapts according to application state.

No Dataset

↓

Import Dataset CTA

Dataset Available

↓

Run Analysis

Analysis Complete

↓

AI Opportunities

Campaign Running

↓

Live Campaign Progress

Stage 4 — Customer Intelligence Workspace

Purpose

Allow marketers to investigate customer segments before launching campaigns.

This page combines

- Customer management
- Customer segmentation
- AI analysis
- Segment exploration

instead of separating them across multiple pages.

Sections

Dataset Summary

↓

Filters

↓

Customer Table

↓

Segment Analysis

↓

AI Recommendations

↓

Create Campaign

Once the marketer understands the audience, they proceed to campaign creation.

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6. Customer Intelligence Analysis

Selecting a customer segment reveals a detailed AI analysis.

Sections

Executive Summary

- Customers

- Revenue Opportunity
- AI Confidence
- Priority

Customer Statistics

- Average Spend
- Average Order Value
- Average Orders
- Average Inactivity

Behavioral Analysis

- Preferred Category
- Preferred Channel
- Active Hours

Business Opportunity

- Expected Revenue
- Expected Orders
- Recovery Probability

AI Recommendations

- Best Channel
- Best Time
- Best Offer
- Best Tone

This page answers

"Why should I target this audience?"

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7. Campaign Workspace

Purpose

Execute campaigns.

The audience has already been selected.

Therefore the campaign page focuses only on execution.

Sections

Campaign Information

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Channel Selection

↓

Message Generation

↓

Media Attachments

↓

Live Preview

↓

Launch Campaign

The page should never ask the marketer to reselect the audience.

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8. AI Message Generation

Users may choose

Option 1

Generate AI Messages

or

Option 2

Write Custom Message

AI Configuration

- Tone
- Brand Voice
- Offer Type

- CTA

Generated content remains editable.

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9. Communication Preview

The preview panel updates live.

Supported previews

- WhatsApp
- SMS
- Email
- RCS

Whenever the marketer changes

- Channel
- Message
- Image
- CTA

the preview updates instantly.

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10. Campaign Execution

Launching a campaign starts the simulation.

Channel Service

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Delivery

↓

Read

↓

Click

↓

Order

↓

Webhook Callback

↓

MongoDB

↓

Frontend Update

The marketer sees campaign progress rather than individual customer events.

Metrics

Queued

Sent

Delivered

Opened

Clicked

Failed

Orders

Revenue

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11. Campaign Analytics

After completion

The campaign moves to Analytics.

Metrics

Recipients

Sent

Delivered

Opened

Read

Clicked

Failed

Orders Generated

Revenue Generated

Conversion Rate

CTR

Delivery Rate

Timeline

Revenue Attribution

Campaign Comparison

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12. AI Responsibilities

Machine Learning

- Customer segmentation
- Customer personas
- Revenue estimation
- Opportunity detection

LLM

- Campaign recommendations
- Marketing copy
- Communication suggestions
- Campaign reasoning

The AI assists decision making.

It never replaces the marketer.

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13. Application States

State 1

No Dataset

Dashboard

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Import Dataset

State 2

Dataset Imported

↓

Run Analysis

State 3

Analysis Complete

↓

Dashboard Recommendations

↓

Customer Intelligence

State 4

Campaign Created

↓

Campaign Execution

State 5

Campaign Completed

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Campaign Analytics

This ensures every page behaves according to available data.

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14. System Architecture

React Frontend

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Express Backend

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MongoDB

↓

Python FastAPI

↓

RFM Analysis

↓

KMeans Clustering

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Customer Personas

↓

LLM

↓

Campaign Generation

↓

Channel Stub

↓

Webhook

↓

Analytics

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15. Design Principles

Every page should answer one business question.

Dashboard

"What should I do today?"

Customer Intelligence

"Who should I target and why?"

Campaign

"How should I communicate?"

Analytics

"Did the campaign succeed?"

No feature should duplicate another page's responsibility.

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16. User Journey

Login

↓

Import Dataset

↓

Automatic AI Analysis

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Dashboard Recommendations

↓

Customer Intelligence

↓

Review Segment

↓

Create Campaign

↓

Generate AI Content

↓

Preview Communication

↓

Launch Campaign

↓

Campaign Progress

↓

Campaign Analytics

↓

Dashboard Updated

This creates a single continuous workflow instead of multiple disconnected pages.

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17. Future Scope

- Predictive churn modeling
- Dynamic audience refresh
- A/B campaign testing
- Multi-brand dataset support
- Real communication providers
- Customer lifetime value prediction
- Campaign scheduling
- Reinforcement learning for campaign optimization